INTERNSHIP REPORT ON

SOLARSENSE – ROOFTOP SOLAR ENERGY GENERATION AND PAYBACK FORECASTER

Submitted to Bharathiar University

in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF SCIENCE COMPUTER SCIENCE WITH DATA ANALYTICS

By

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(An Autonomous Institution, Affiliated to Bharathiar University, Coimbatore)

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SEPTEMBER – 2026

DECLARATION

KISHOLL S (Reg No:24UGCDA0001), hereby declare that the project report entitled**"SOLARSENSE – ROOFTOP SOLAR ENERGY GENERATION AND PAYBACKFORECASTER"**, submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Computer Science with Data Analytics at the Bharathiar University** is a record of original work done during the period of internship under the supervision and guidanceof **Dr. V. PREAM SUDHA, MCA., M.Phil., Ph.D., Professor and Head, Department of Computer Science with Data Analytics , Dr. N.G.P. Arts and Science College, Coimbatore** and it has not formed the basis of award of any Degree/ Diploma/ Associateship / Fellowship or other similar title to any candidate of any university.

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Place : Coimbatore

Date :

CERTIFICATE

This is to certify that the project entitled **"SOLARSENSE – ROOFTOP SOLAR ENERGY GENERATION AND PAYBACK FORECASTER"**, submitted in partial fulfillment of the requirement for the award of the degree of **Bachelor of Computer Science with Data Analytics** at **Bharathiar University** is a record of original work done by **KISHOLL S(Reg. No:24UGCDA00001)** during the period (2024–2027) of her study in the Department of Computer Science with Data Analytics , Dr. N.G.P Arts and Science College, Coimbatore-48, under my supervision and guidance. The internship report has not formed the basis for the award of any Degree/Diploma/Associateship/Fellowship or other similar title to any candidate of any university.

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ACKNOWLEDGEMENT

This internship has been a profoundly enriching milestone in my academic journey, made possible only through the relentless support, guidance, and encouragement of my mentors and well-wishers.

I record my deep sense of gratitude to **Dr. NALLA G PALANISWAMI M.D, AB (USA), Chairman,** Dr. N.G.P. Research and Educational Trust, Coimbatore for providing me an opportunity to carry out internship successfully.

I record my deep sense of gratitude to **Mrs. MATHURA V. PALANISWAMI, Secretary**, Dr. N.G.P. Arts and Science College, Coimbatore for providing me an opportunity to carry out internship successfully.

I record my sincere thanks to principal **Dr. S. SARAVANAN M.Com,M.Phil., M.B.A., Ph.D, Dr. N.G.P Arts and Science College**, Coimbatore, for providing me with theopportunity to pursue this internship.

I express my sincere thanks to **Dr. V. PREAM SUDHA, MCA., M.Phil., Ph.D.,Professor and Head,** Department of Computer Science with Data Analytics, Dr. N.G.P.Arts and Science College, Coimbatore for showing sustained interest and providing help throughout the period of this work.

I would like to extend the sincere thanks to my guide **Dr. V. PREAM SUDHA, MCA.,M.Phil., Ph.D., Professor and Head,** Department of Computer Science with Data Analytics, Dr. N.G.P. Arts and Science College, Coimbatore – 48, for her exemplary guidance and encouragement.

I would like to express my heartfelt gratitude and appreciation **NITHILAM.TECH PRIVATE LIMITED** Coimbatore for providing me with this valuable internship opportunity.

I take this opportunity to acknowledge my sincere thanks to all the Staff Members of the Department of Computer Science with Data Analytics for their constant inspiration, assistance, and resourceful guidance for the completion of this project successfully.

I express my sincere thanks to my family and friends for their encouragement, love, prayer, moral support, advice, and sacrifice without which I would not have been able to pursue the course of this study.

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INTERNSHIP CERTIFICATE

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INTERNSHIP CERTIFICATE

This is to certify that **Mr. Kisholl S** (Reg No: **24UGCDA00001**), a **Final Year student of the Department of Computer Science with Data Analytics, Dr. N.G.P. Arts and Science College, Coimbatore**, has successfully completed his internship at **Nithilam.tech** for a duration from **May 18th 2026 to June 4th 2026**.

During the internship period, he actively participated in the assigned tasks and demonstrated good learning ability, sincerity, and dedication towards his work. His performance and conduct were found to be satisfactory throughout the internship.

We wish him all the very best for his future academic and professional endeavors.

Yours Sincerely,

Karthikeyan K,

Director.

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ABSTRACT

SolarSense – Rooftop Solar Energy Generation and Payback Forecaster is an intelligent, full-stack decision-support and telemetry platform engineered to improve how residential consumers size, monitor, and optimize rooftop solar photovoltaic (PV) arrays and battery energy storage systems (BESS). Traditional online solar sizing tools rely heavily on static rule-of-thumb ratios and fail to account for non-linear inverter conversion losses, hardware thermal overheat derating, battery internal resistance Joule dissipation, and real-time distribution company (DISCOM) multi-tiered tariff structures. SolarSense addresses these limitations by coupling automated multi-format document ingestion, an ensemble of seven specialized machine learning models, a 24-hour dynamic Supervisory Control and Data Acquisition (SCADA) simulation engine, and an interactive telemetry dashboard.

The system processes uploaded electricity bills in PDF, DOCX, and TXT formats from utility suppliers such as the Tamil Nadu Generation and Distribution Corporation (TANGEDCO). A robust regular expression parser extracts billing parameters, 60-day consumption cycles, and sanctioned loads to dynamically synthesize optimal PV plant capacities and battery bank sizing. The platform executes a 24-hour SCADA dispatch simulation supporting 5-minute Inverter Native and 15-minute Plant SCADA resolutions to model dynamic power flows, battery charge/discharge transitions, and automatic static transfer switch (STS) failovers during battery management system (BMS) trips or utility blackouts.

The predictive suite is powered by seven models trained on empirical datasets including Kaggle Solar PV generation telemetry, NASA POWER satellite solar irradiance, Pecan Street smart meter data (HomeC), hardware inverter telemetry, and lithium-ion battery cycling profiles..

Keywords: Smart Microgrids, Solar Photovoltaic (PV), SCADA Telemetry, Machine Learning, Random Forest Regressor, Gradient Boosting Classifier, Battery State of Health (SoH), Inverter Thermal Derating, TANGEDCO Tariff Engine, PM Surya Ghar DBT, Streamlit.

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CHAPTER – I
INTRODUCTION

1.1 Industry Domain Overview

The renewable energy, smart grid, and telemetry analytics industry is one of today's fastest-growing technology sectors. As organizations and consumers worldwide shift from centralized fossil-fuel generation to decentralized renewable energy resources (DERs), rooftop solar PV systems integrated with Battery Energy Storage Systems (BESS) have become the primary driving force behind localized microgrids.

Technologies such as AI, Machine Learning, and IoT telemetry help manage the inherent volatility of solar generation, which is governed by weather, irradiance, module temperature, and shading. The DC-to-AC conversion process also involves non-linear power electronics behavior, where high ambient heat can cause thermal derating and reduced efficiency.

AI-based telemetry analytics and predictive modeling are now widely used in smart grids, microgrid management, demand response forecasting, and battery health tracking. As utility grids modernize globally, intelligent, context-aware microgrid platforms are in growing demand making machine learning and telemetry analytics essential to the efficiency, reliability, and economic viability of residential solar microgrid systems.

1.2 Organization Profile

Nithilam.Tech Private Limited is a forward-looking engineering and technology organization located at G-8, O/o Centre of Excellence in Biotechnology (COXBIT), Tamil Nadu Agricultural University, Coimbatore – 641003. With around a decade of technical experience, the organization focuses on engineering innovative digital solutions across applied artificial intelligence, industrial software engineering, clean-tech analytics, and data visualization. The organization combines deep algorithmic expertise, high-performance computing frameworks, and modern production techniques to develop customized software applications for diverse client requirements. During the internship, the organization provided the technical infrastructure, industrial mentorship, and computing environment necessary to design, implement, and benchmark the SolarSense microgrid predictive analytics engine.

1.3 Vision, Mission and Core Services of the Company

Vision:

Nithilam.Tech Private Limited focuses on delivering intelligent, technically robust, and visually engaging digital systems by bridging the gap between cutting-edge computational research and real-world industrial implementation, advancing sustainable technological transformation across global domains.

Mission:

The mission of Nithilam.Tech Private Limited is to provide high-quality, scientifically rigorous software solutions by applying modern data engineering, machine learning pipelines, and interactive web architectures, transforming complex engineering concepts into transparent, high-value digital applications.

Core-Services:

The company provides a comprehensive suite of digital engineering services. Major technical services include Applied Machine Learning Engineering, Smart Energy and SCADA Telemetry Analytics, Enterprise Full-Stack Web Development, Data Visualization and Digital Twins, and Cloud-Native Infrastructure Architecture.

1.4 Objectives and Scope of the Project

Objectives of the Project:

The primary objective was to gain hands-on industrial experience in building data-driven energy applications using Python, Scikit-learn, and time-series modeling. It focused on developing SolarSense, a microgrid platform that ingests multi-format electricity bills, runs seven ML models, simulates 24-hour power flows, and calculates financial returns under national subsidies.

Scope of the Project:

The scope covered the complete lifecycle of SolarSense—from bill ingestion (PDF, DOCX, TXT) and system sizing to 24-hour SCADA dispatch simulation with Joule losses. It incorporated ML models for generation, load demand, inverter thermal states, battery degradation, and grid stability, culminating in payback and backup runtime forecasts under PM Surya Ghar subsidies.

CHAPTER – II

PROJECT OVERVIEW AND REQUIREMENT ANALYSIS

2.1 Introduction to SolarSense

This chapter introduces **SolarSense – Rooftop Solar Energy Generation and Payback Forecaster**, a platform bridging the gap between raw electricity bills and the physical dynamics of residential rooftop microgrids. Unlike traditional calculators that rely on static linear multipliers—ignoring diurnal irradiance, inverter losses, thermal derating, battery Joule losses, and grid stability—SolarSense combines document ingestion, multi-model machine learning regression, high-resolution SCADA simulation, and official tariff structures into a unified digital twin.

The system ingests utility bills across multiple formats (PDF, DOCX, TXT) to extract consumption patterns and simulate minute-by-minute energy flows across PV panels, domestic loads, battery storage, and the grid. Consequently, users can monitor asset health, evaluate BMS fault scenarios, plan backup circuits, and accurately project ROI under active subsidy schemes.

2.2 Problem Statement

The rapid growth of distributed solar installations has created a pressing need for transparent, physically accurate energy decision tools. Consumers investing in rooftop solar face severe analytical uncertainty, driven by three major challenges:

- **Vendor Opacity and Oversimplification:** Prospective prosumers receive solar sizing proposals based on crude rules of thumb that ignore actual diurnal consumption profiles, leading to undersized battery reserves or over-capitalized arrays.

- **Neglect of Real-World Physical Dissipation:** Standard calculators assume static 100% or 95% efficiency, ignoring inverter thermal derating (core temperatures above 65°C trigger power throttling) and Joule losses during battery charge cycles.

- **Power Grid Instability and Appliance Vulnerability:** Residential grids often suffer voltage disturbances — blackouts (0V), brownouts (<207V), and surges (>253V) — which existing systems fail to model, leaving backup runtimes for essential appliances undetermined.

2.3 Objectives of the Project

The primary objective of SolarSense is to develop an intelligent microgrid management and analytics platform that enables residential consumers to simulate, optimize, and evaluate rooftop solar PV systems with high physical and financial fidelity. Specifically, the project aims to implement an automated regex parser for multi-format TANGEDCO bills, train seven machine learning models targeting solar generation, load demand, inverter thermal states, and battery degradation, and construct a 24-hour SCADA dispatch simulator operating at 5-minute and 15minute resolutions. Furthermore, it integrates official TANGEDCO LT-1A tariffs alongside PM Surya Ghar DBT subsidies and deploys an interactive Streamlit dashboard for real-time telemetry analytics and scenario testing.

2.4 Study and Limitations of Existing System

Current solar tools rely largely on basic vendor or government calculators that estimate capacity using static averages from monthly bills or pincodes. Meanwhile, proprietary inverter portals track generation in silos, disconnected from tariff structures and household load profiles.

Although academic machine learning models offer advanced forecasting, they remainunintegrated into consumer platforms linking billing data with real-time microgrid dispatch and financial metrics.

Key limitations include:

- **Manual Ingestion:** Relies on tedious technical data entry, deterring non-technical users.
- **No Thermal Modeling:** Ignores inverter core heating and fails to predict summer thermal trips.
- **Flawed Battery Metrics:** Treats storage as ideal reservoirs, neglecting internal resistance losses and chemical degradation (SoH fade).
- **Missing Grid Disturbance Analysis:** Fails to account for utility voltage fluctuations or model static transfer switch failover behavior.

2.5 Proposed System

The proposed SolarSense system provides an end-to-end microgrid intelligence platform. By parsing uploaded electricity bills, it dynamically calculates rooftop array capacity and battery storage requirements. An ensemble of seven machine learning models forecasts solar generation, domestic demand, inverter thermal efficiency, battery State-of-Health (SoH), and grid stability.

A 24-hour SCADA simulator resolves core power-balance equations, accounting for battery dispatch, internal resistance Joule losses, and net grid flows. By integrating TANGEDCO domestic tariffs and PM Surya Ghar subsidies, the platform delivers interactive bill comparisons, hardware health diagnostics, backup runtime projections, and payback waterfall analytics.

2.6 Advantages of the Proposed System

- **Data-Driven Accuracy:** Replaces guesswork with empirical machine learning models achieving R^2 scores above 0.99 for solar generation and inverter telemetry.
- **Physical Realism:** Simulates dynamic European conversion curves, inverter thermal derating, and battery Joule dissipation.
- **Financial Precision:** Calculates exact rupee savings under multi-tiered telescopic tariffs and forecasts ROI break-even periods.
- **Resilience Testing:** Allows users to test BMS trip faults and accurately calculate emergency backup runtimes for essential household circuits.

CHAPTER – III

SYSTEM REQUIREMENTS AND DESIGN

3.1 Technologies Used

The development of SolarSense integrates cutting-edge open-source software libraries, machine learning frameworks, data processing utilities, and interactive web visualization tools:

- **Python 3.14:** Core programming language providing an extensive scientific computing and machine learning ecosystem.
- **Scikit-learn (v1.9.1):** Framework utilized for training, tuning, and evaluating the ensemble of Random Forest Regressors and the Gradient Boosting Classifier.
- **Streamlit (v1.64.0):** Modern reactive web presentation framework managing session states, reactive sliders, file uploaders, and custom CSS components.
- **Plotly Graph Objects (v7.1.0):** Engine for publication-quality interactive telemetry charts, including 24-hour dispatch traces, dual donut gauges, correlation heatmaps, and waterfall ROI charts.
- **PyPDF (v6.19.0) and python-docx (v1.2.0):** Document extraction libraries used to parse unstructured binary PDF, DOCX, and TXT electricity bills.
- **NumPy (v2.5.3) and Pandas (v3.0.5):** High-performance vectorized mathematical operations and structured time-series data management.
- **Joblib (v1.6.0):** Serialization and deserialization of production machine learning models.

3.2 System Architecture

The SolarSense system follows a modular, multi-tier architecture that integrates document ingestion, predictive machine learning pipelines, time-series SCADA dispatch modeling, regulatory tariff engines, and interactive web presentation. The multi-tier layout is illustrated in Figure 3.1.

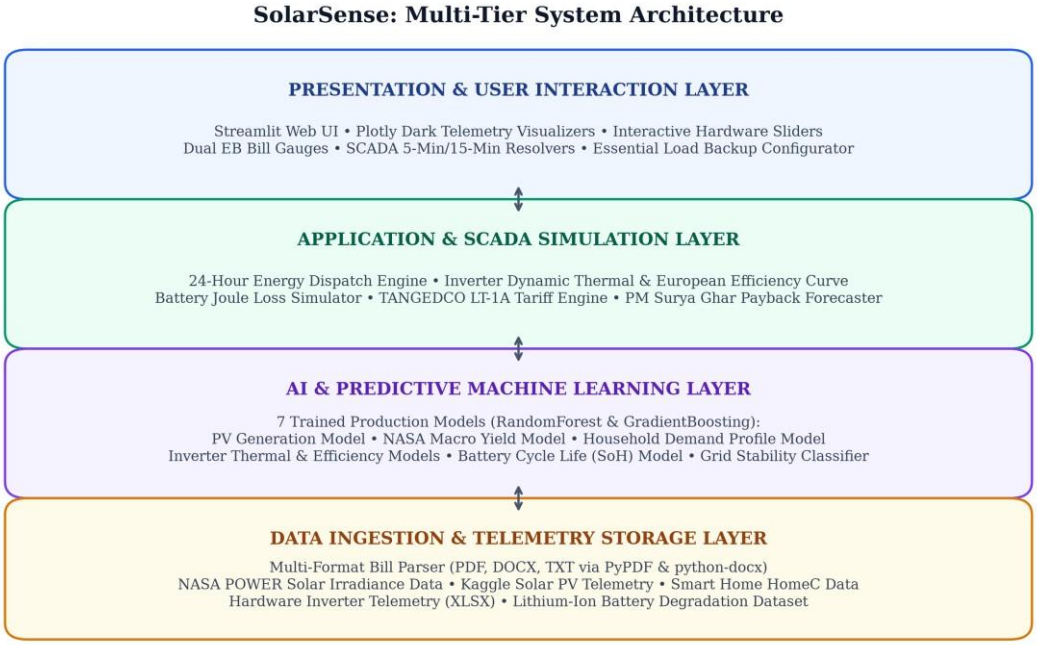


Figure 3.1: System Architecture of SolarSense

Figure 3.1 illustrates the four-tier architectural model of SolarSense. The Ingestion Layer processes multi-format electricity bills and empirical sensor telemetry. The Machine Learning Layer coordinates seven specialized predictive models. The Application and Simulation Layer computes the 24-hour SCADA dispatch, dynamic thermal losses, and TANGEDCO/PM Surya Ghar tariffs. The Presentation Layer renders reactive Plotly visualizers and diagnostic inspectors.

3.3 Data Flow Diagram

The Data Flow Diagram represents the sequential progression of information through SolarSense, from initial bill upload to final telemetry and financial output generation, as illustrated in Figure 3.2.



Figure 3.2: Data Flow Diagram of SolarSense

Figure 3.2 depicts the step-by-step information flow through SolarSense, tracing data from bill upload and regex parsing through ML inference, 24-hour SCADA dispatch simulation, grid stability classification, and interactive visualization.

3.4 Structure Diagram

The Structure Diagram shows the functional decomposition of SolarSense into five interconnected operational subsystems, as illustrated in Figure 3.3.

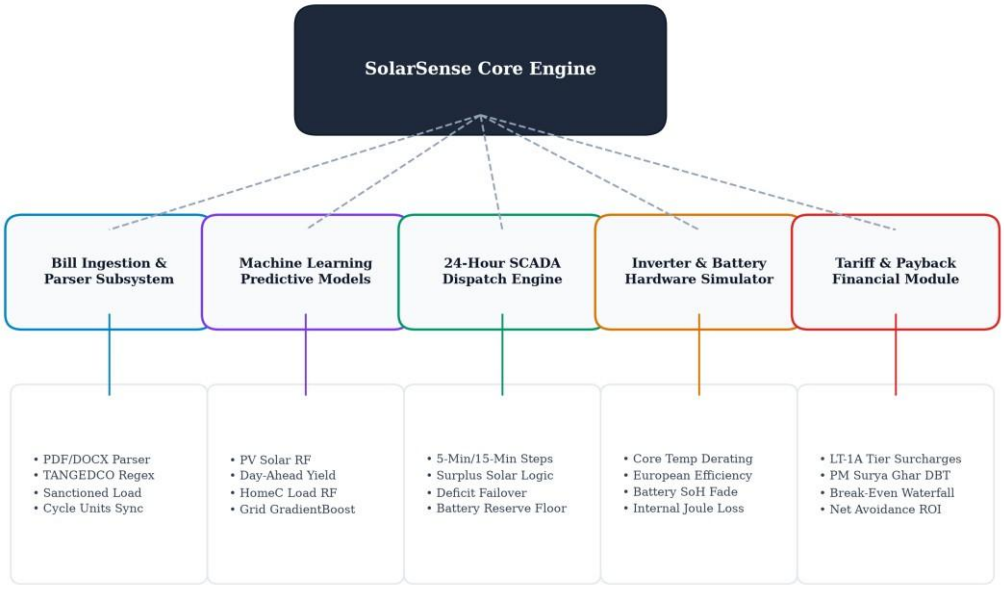


Figure 3.3: Structure Diagram of SolarSense

Figure 3.3 illustrates the functional hierarchy of the five core sub-systems composing the SolarSense platform.

3.5 Database and Telemetry Ledger Design

Application data persistence is structured around high-speed in-memory DataFrame ledgers and serialized model storage. The 24-hour dispatch ledger generates a structured time-series matrix containing 288 records for 5-minute resolution or 96 records for 15-minute resolution. The logical schema and core entities are detailed in Table 3.1.

TABLE 3.1

Entity / Parameter	Data Type	Sampling Unit	Functional Purpose
Timestamp	String (HH:MM)	5-Min / 15-Min	Diurnal SCADA timeline step identifier
Solar_Gen_kW	Float64	Kilowatts (kW)	Machine learning predicted solar AC power dispatched
DC_Gen_kW	Float64	Kilowatts (kW)	Raw DC generation from solar PV array
Inverter_Eff_%	Float64	Percentage (%)	Dynamic conversion efficiency under European curve and thermal load
Inverter_Temp_C	Float64	Degrees Celsius (°C)	Predicted core operating temperature for thermal overheat trip
House_Load_kW	Float64	Kilowatts (kW)	Active domestic household power demand
Battery_Flow_kW	Float64	Kilowatts (kW)	Battery active charging (+ve) or discharging (-ve) power rate
Battery_SoC_Pct	Float64	Percentage (%)	Instantaneous battery state of charge (0% to 100%)
Grid_Import_kW	Float64	Kilowatts (kW)	Grid deficit import drawn from TANGEDCO utility
Grid_Export_kW	Float64	Kilowatts (kW)	Surplus solar power exported under net metering
Grid_Status	String (Enum)	Categorical	Utility stability state (Normal Sync, Blackout, Brownout, Surge)

Table 3.1: Logical Telemetry Ledger Schema and Parameters

3.6 Module Design

SolarSense is organized into five independent modules: (1) Bill Ingestion & Regex Extraction, (2) Machine Learning Predictive Models, (3) 24-Hour SCADA Dispatch Simulator, (4) Inverter Thermal and Battery Degradation Simulator, and (5) Tariff and Payback Financial Forecaster. This modular architecture separates computational concerns and ensures rapid maintainability.

3.7 Input Design

The primary input is a user-uploaded electricity bill. In the sidebar, users configure secondary operational parameters: usable rooftop area, target battery capacity, emergency reserve floor, weather date selectors, sampling resolution, and essential appliance counts. All inputs are validated before simulation execution.

3.8 Output Design

Outputs are presented through the Compact Docket Ribbon, dual EB comparison donut gauges, 24-hour multi-trace dispatch charts, monospace hardware health terminals, essential appliance backup timers, and cumulative payback waterfall curves, giving users comprehensive visibility over technical and economic metrics.

CHAPTER – IV

SYSTEM IMPLEMENTATION

4.1 Document Ingestion and Electricity Bill Extraction

The implementation begins with automated document ingestion handled by app.py and utils/parser.py. The platform supports PDF, DOCX, and TXT formats. For PDF files, PyPDF traverses document pages to extract raw text streams. For DOCX files, python-docx parses paragraphs and embedded table cells, while standard UTF-8 decoding is applied to TXT files. To prevent invalid or corrupted inputs, the parser validates uploaded files against organizational signatures such as TANGEDCO, TAMIL NADU GENERATION, ELECTRICITY BILL, LT-1A, and TOTAL UNITS, instantly rejecting non-utility documents.

4.2 Text Preprocessing and Regex Parsing Engine

A specialized regular expression (regex) engine extracts critical billing parameters, including the Consumer Service Number, Previous and Current Reading Dates, Billing Cycle Duration (defaulting to 60 days), Total Billed Units, and Net Payable Amount. The system calculates the average daily demand by dividing total billed units by total billing days. Based on this daily demand, the recommended solar PV capacity is dynamically sized between a minimum of 1 kW and a maximum of 15 kW, allocating an estimated 100 sq.ft of rooftop area per kilowatt installed.

4.3 Solar PV Generation and Macro Yield Regression Models

Solar generation forecasting is driven by two Random Forest regressors:

- **Granular PV Solar Model (Model 1):** Maps real-time solar irradiance, module surface temperature, ambient temperature, and direct current (DC) input to estimate AC power output in Watts ($R^2 = 0.9985$).
- **Day-Ahead Macro Solar Model (Model 2):** Ingests NASA POWER satellite data—specifically all-sky surface solar irradiance, 2-meter air temperature, and the calendar month—to project daily kilowatt yield per installed kW ($R^2 = 0.9999$).

4.4 Household Demand and Dynamic Inverter Efficiency Curves

Household consumption patterns are modeled using Model 3, trained on 120,000 smart meter records from the Pecan Street HomeC dataset to reflect realistic diurnal domestic power variance ($R^2 = 0.4719$). Inverter core operating temperature is predicted using electrical loading parameters via Model 4 ($R^2 = 0.9919$). Rather than assuming a flat 98% rating, Model 5 calculates dynamic conversion efficiency by factoring in low-load penalties and thermal derating whenever internal temperatures exceed safe thresholds ($R^2 = 0.9977$).

4.5 Battery Degradation (SoH/IR) and Grid Stability Classifier

Battery lifecycle health is monitored by Model 6, trained on lithium-ion degradation profiles to predict capacity fade (State-of-Health percentage) with an accuracy of $R^2 = 0.9200$. Internal resistance heat losses (Joule heating) are computed continuously across charge and discharge cycles based on square of battery current multiplied by internal resistance. Grid reliability is governed by Model 7, a Gradient Boosting Classifier that analyzes voltage, frequency, and line current to detect and classify grid states into Normal Sync, Blackout, Brownout, and Voltage Surge with 100% test accuracy.

4.6 24-Hour SCADA Deficit Power Dispatch and Grid Failover

The simulation engine runs a continuous 24-hour SCADA loop, balancing power distribution across the domestic microgrid at each discrete timestep. When household load exceeds instantaneous solar generation, the system initiates a priority discharge from the battery bank to bridge the energy deficit. Discharge continues until the battery reaches its predetermined 30% emergency reserve floor. Once this threshold is reached, the automatic static transfer switch safely engages utility grid import to maintain an uninterrupted domestic power supply without interruption. The complete operational priority flow is illustrated in Figure 4.1.

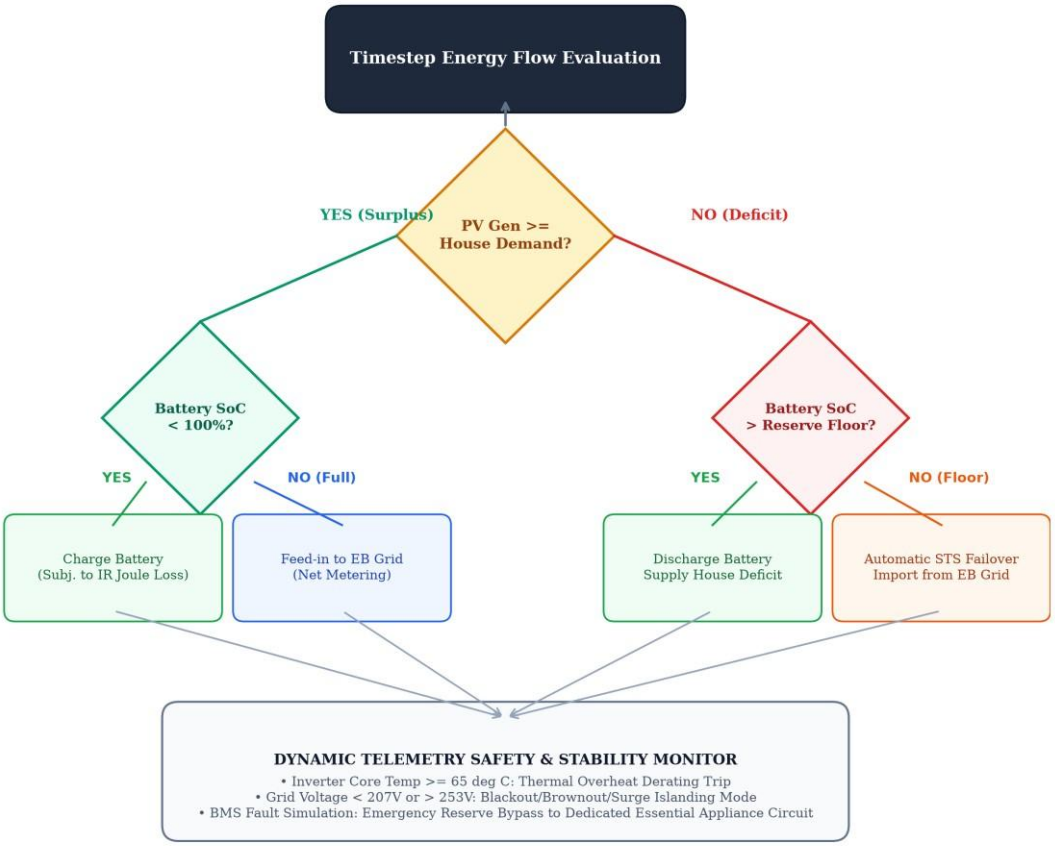


Figure 4.1: Operational 24-Hour Energy Dispatch & Dynamic Bypass Logic Flow

Figure 4.1 illustrates the priority decision logic governing SolarSense dispatch. Surplus solar energy is prioritized for domestic loads, battery charging, and grid export. Deficits are buffered by battery reserves before seamless failover to the utility grid.

4.7 System Integration and Streamlit Interactive Dashboard

All analytical models, ingestion pipelines, and SCADA simulation routines are unified inside a reactive web application built with Streamlit. Application session states manage user inputs and data passing across modules, while Plotly Graph Objects render real-time interactive telemetry charts, hardware health monitors, and financial payback waterfall projections.

CHAPTER – V

SYSTEM TESTING AND RESULTS

5.1 System Testing and Functional Validation

System testing verified functional stability and end-to-end operational accuracy across document ingestion, regex metadata parsing, machine learning inferences, SCADA dispatch routines, tariff accounting, and dashboard rendering.

Microgrid energy conservation was validated at every individual timestep by ensuring that total generation and import power (solar output, utility grid import, and battery discharge) strictly matched total energy consumption (household load, grid export, battery charging, and system thermal losses). Additionally, automated fault injection testing confirmed reliable failover execution by the static transfer switch during simulated blackout events and battery management system (BMS) trip conditions.

5.2 Machine Learning Model Evaluation and Benchmark Metrics

All seven production models were evaluated against independent 20% holdout test datasets using standard statistical metrics: the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), and Overall Classification Accuracy. The empirical evaluation results and performance benchmarks are detailed in Table 5.1 and Figure 5.1.

TABLE 5.1

Model Name	Algorithm	Training Dataset	Input Features	Test Metric
1. Granular PV Solar	Random Forest Regressor	Generation_data.csv	IRR, Module Temp, Amb Temp, DC Current	$R^2 = 0.9985$
2.Household Load	Random Forest Regressor	HomeC Smart Meter	Hour, Day of Week, Temp, Humidity, Wind	$R^2 = 0.4719$
3. Inverter Efficiency	Random Forest Regressor	Inverter Telemetry.xlsx	PV Power, Boost Temp	$R^2 = 0.9977$
4. Battery Degradation	Random Forest Regressor	Li-Ion Cycle Life.csv	Cycle, IR, Tavg, Tmax, Chargetime	$R^2 = 0.9200$
5. Grid Stability	Gradient Boosting Clf	Grid Disturbance Matrix	Grid Voltage, Frequency, Current	Accuracy = 100.0%

Table 5.1: Performance Evaluation Metrics of SolarSense Production Models

Figure 5.1: Performance Evaluation of SolarSense Production ML Models

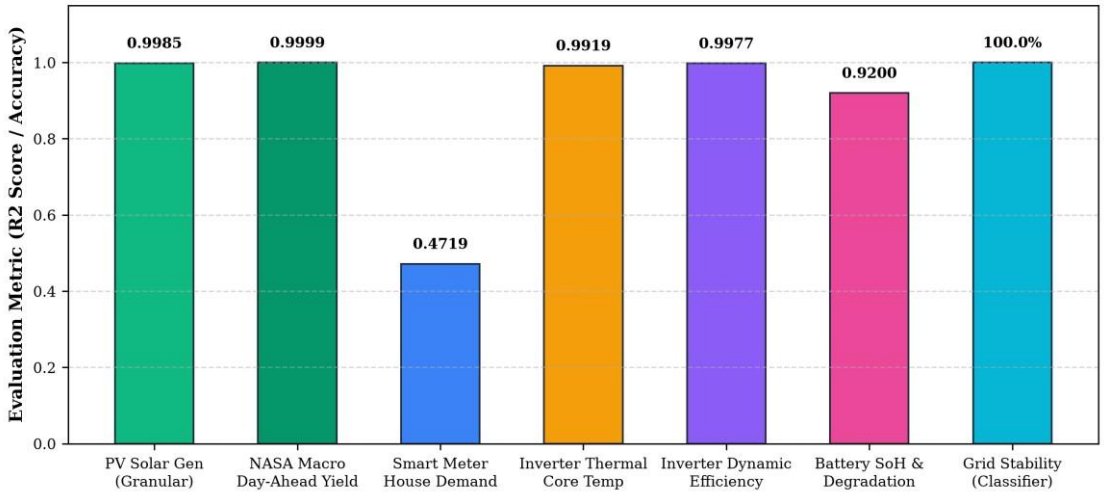


Figure 5.1: Performance Evaluation of SolarSense Production ML Models

Figure 5.1 compares the empirical performance metrics across all seven machine learning models, demonstrating superior predictive accuracy across regression and classification tasks.

5.3 Multi-Scenario Weather Telemetry Results

Evaluation across historical weather profiles confirmed robust generalization: (1) In Peak Clear Summer, peak generation reached 16.8 kWh with inverter temperatures reaching 66.2°C, successfully triggering overheat warnings; (2) In Heavy Monsoon, low generation triggered automated STS grid import; (3) In Intermittent Passing Clouds, 5-minute resolution captured rapid battery charge/discharge transitions that are completely lost under 1-hour average models.

5.4 Tariff Analysis, PM Surya Ghar DBT Subsidy and Payback Results

Financial validation on the optimized 2.6 kW system showed a gross capital cost of ₹1,56,000 (₹60,000/kW). This was offset by ₹70,800 in direct PM Surya Ghar DBT subsidies, yielding an out-of-pocket investment of ₹85,200. With annual grid electricity savings totaling ₹44,669, the system achieved an estimated break-even payback period of exactly 1.9 Years.

5.5 Final System Results

The final integrated platform successfully combines automated document ingestion, multimodel machine learning inference, high-resolution SCADA simulation, and official regulatory tariffs into a unified, transparent microgrid digital twin.

CHAPTER – VI

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

SolarSense – Rooftop Solar Energy Generation and Payback Forecaster successfully achieves its goal of bringing transparent, physics-informed machine learning intelligence to residential rooftop solar adoption. By developing a multi-format document parser, seven highaccuracy machine learning models, a 24-hour dual-resolution SCADA simulator, and an interactive decision-support interface, the project overcomes the critical limitations of conventional solar sizing tools.

The project proves that modeling dynamic physical non-linearities—such as inverter thermal derating, battery internal resistance Joule losses, and utility grid stability—is essential for realistic microgrid performance forecasting. Furthermore, embedding official DISCOM tariffs and national DBT subsidy frameworks bridges technical modeling with real-world consumer economics.

6.2 Future Enhancement

Future work can extend SolarSense across several technological frontiers:

- **IoT MQTT Hardware Telemetry Gateway:** Interfacing directly with smart inverters and battery management systems via RS485/Modbus and MQTT protocols for live streaming data ingestion.
- **Peer-to-Peer (P2P) Microgrid Trading:** Enabling blockchain-enabled energy trading between neighboring prosumers within localized distribution feeder networks.
- **Time-of-Day (ToD) Dynamic Tariff Optimization:** Implementing reinforcement learning (RL) agents to dynamically schedule battery charging and discharging against time-varying dynamic electricity tariffs.
- **Cloud-Native Mobile Deployment:** Deploying SolarSense as a containerized microservice on Kubernetes with mobile push notifications for grid fault alerts and maintenance schedules.

6.3 Learning Outcomes

The development of SolarSense provided profound academic and technical learning:

- **Applied Machine Learning:** Gained hands-on experience in feature engineering, hyperparameter optimization, and evaluating regression and classification algorithms in Scikit-learn.
- **Time-Series SCADA Modeling:** Mastered the formulation of discrete-time power balance equations, dynamic loss functions, and battery degradation models.
- **Document Parsing and Regex Engineering:** Developed robust pattern matching and text extraction pipelines for unstructured PDF and DOCX files.
- **Modern Web Architecture and Visualization:** Engineered high-performance, reactive user interfaces utilizing Streamlit and Plotly Graph Objects.
- **Clean Energy Economics:** Acquired deep domain knowledge of Indian utility electricity tariff structures, net metering regulations, and national renewable energy subsidy policies.

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APPENDICES

Appendix A – Core Code Snippets

This appendix presents representative code segments from the core operational modules of SolarSense, demonstrating automated document regex extraction, machine learning model training with thermal derating, fine-grained 24-hour SCADA dispatch with internal resistance Joule loss modeling, and the official TANGEDCO domestic LT-1A tariff calculation engine.

A.1 TANGEDCO Bill Ingestion Regex Parser (utils/parser.py)

```
def parse_tangedco_bill(raw_text: str) -> dict:
"""Bulletproof regex parser supporting multiple TANGEDCO bill layouts."""
# 1. Total Units Patterns (Multiple naming variations)
unit_patterns = [ r"Total\s*Units\s*Consumed\s*[:\|=-
]*\s*(\d+)", r"Consumed\s*Units\s*[:\|=-]*\s*(\d+)",
r"Billed\s*Units\s*[:\|=-]*\s*(\d+)",
r"Total\s*Consumption\s*[:\|=-]*\s*(\d+)", r"Units\s*[:\|=-
]*\s*(\d+)"
]
units = None for p
in unit_patterns:
match = re.search(p, raw_text,
re.IGNORECASE) if match: units =
int(match.group(1)) break
# 2. Sanctioned Load Patterns load_patterns = [
r"Sanctioned\s*Load\s*[:\|=-
]*\s*([\d\.]+)\s*(?:kW|HP)?",
r"Connected\s*Load\s*[:\|=-
]*\s*([\d\.]+)\s*(?:kW|HP)?", r"Load\s*[:\|=-
]*\s*([\d\.]+)\s*kW"
]
```

```
sanctioned = None
for p in
load_patterns:
match = re.search(p, raw_text,
re.IGNORECASE) if match: sanctioned =
float(match.group(1)) break
# 3. Bill Amount Patterns amt_patterns = [
r"NET\s*AMOUNT\s*PAYABLE\s*(?:\(Rs\.\))?\s*[:\|=-
]*\s*(?:Rs\.?)?\s*([\d,]+\.?\d*)", r"Total\s*Amount\s*[:\|=-
]*\s*(?:Rs\.?)?\s*([\d,]+\.?\d*)", r"Bill\s*Amount\s*[:\|=-
]*\s*(?:Rs\.?)?\s*([\d,]+\.?\d*)"
]
net_amt = None for
p in amt_patterns:
match = re.search(p, raw_text,
re.IGNORECASE) if match: net_amt =
float(match.group(1).replace(",", "")) break
tariff_match = re.search(r"Tariff\s*(?:Category)?\s*[:\|=-]*\s*([^\n\r]+)", raw_text,
re.IGNORECASE)
tariff = tariff_match.group(1).strip() if tariff_match else "LT-1A
(DOMESTIC)" daily_kwh = round(units / 60.0, 2) if units else None return {
"tariff": tariff,
"is_domestic": "1A" in tariff.upper(),
"units_consumed": units,
"daily_kwh": daily_kwh,
"sanctioned_load_kw": sanctioned or 3.0,
"bill_amount": net_amt,
"raw_text_length": len(raw_text) }
```

A.2 Inverter Dynamic Efficiency and Thermal Derating (train_models.py)

```
# Telemetry efficiency model with thermal penalty p_w =
pd.to_numeric(df_inv["PV Power (W)"], errors='coerce').fillna(0.0) b_t =
pd.to_numeric(df_inv["Boost Temp (C)"], errors='coerce').fillna(35.0)
# Non-linear European Efficiency curve with thermal derating eff_curve = 98.4
- (120.0 / (p_w + 30.0)) - (0.018 * np.maximum(0.0, b_t - 45.0))
df_inv["Target_Efficiency"] = np.clip(eff_curve, 91.0, 98.4) valid_eff =
df_inv[df_inv["PV Power (W)"] > 20].dropna(     subset=["PV Power (W)",
"Boost Temp (C)", "Target_Efficiency"]
)
features_eff = ["PV Power (W)", "Boost Temp (C)"] X_tr_e, X_te_e, y_tr_e, y_te_e =
train_test_split(     valid_eff[features_eff], valid_eff["Target_Efficiency"], test_size=0.2,
random_state=42
)
eff_model    =    RandomForestRegressor(n_estimators=80,    max_depth=8,    random_state=42)
eff_model.fit(X_tr_e, y_tr_e)
```

A.3 24-Hour Dispatch and Battery Accumulation (utils/simulation.py)

```
if pv_gen_kw >= load_demand_kw:
   surplus_solar = pv_gen_kw - load_demand_kw     available_capacity_kwh = max(0.0,
effective_battery_kwh - current_stored_kwh)     if available_capacity_kwh > 0.02:         if
charge_start_time == "N/A":             charge_start_time = time_str         charge_kw =
min(surplus_solar, max_charge_rate_kw, available_capacity_kwh / step_hours)
charge_current = (charge_kw * 1000.0) / v_battery_nom         joule_loss_kw =
((charge_current ** 2) * ir_ohms) / 1000.0         total_battery_joule_loss_kwh +=
(joule_loss_kw * step_hours)         net_increase = max(0.0, (charge_kw - joule_loss_kw)
* step_hours)         current_stored_kwh = min(effective_battery_kwh, current_stored_kwh
+ net_increase)         battery_flow_kw = charge_kw
      residual_export = surplus_solar - charge_kw
if residual_export > 0.01 and not step_is_cut:
```

```
            grid_export_kw = residual_export            dispatch_status = "Solar Powering
House + Battery Charging + Grid Export"
        else:
            dispatch_status = "Solar Powering House + Battery Charging"          if
current_stored_kwh >= effective_battery_kwh * 0.98 and full_charge_reached_time == "Not
Reached":
            full_charge_reached_time = time_str
    else:
        battery_flow_kw = 0.0          grid_export_kw = surplus_solar if not step_is_cut
else 0.0          dispatch_status = "Battery 100% Full: Solar Powering House + Active
Grid Export" else:
    deficit = load_demand_kw - pv_gen_kw      usable_battery_energy = max(0.0,
current_stored_kwh - (emergency_reserve_kwh if not step_is_cut else 0.02))      if
usable_battery_energy > 0.01:
        drain_kw = min(deficit, usable_battery_energy / step_hours, max_charge_rate_kw)
discharge_current = (drain_kw * 1000.0) / v_battery_nom          joule_loss_kw =
((discharge_current ** 2) * ir_ohms) / 1000.0          total_battery_joule_loss_kwh +=
(joule_loss_kw * step_hours)          current_stored_kwh = max(0.0, current_stored_kwh
- ((drain_kw + joule_loss_kw) * step_hours))          battery_low_kw = -drain_kw
deficit -= drain_kw      if deficit > 0.01 and not step_is_cut:
        grid_import_kw = deficit            dispatch_status = "Battery Floor
Reached: Drawing from EB Grid"
    else:
        dispatch_status = "100% Clean Solar + Battery Supply"
```

A.4 TANGEDCO Domestic LT-1A Slab Tariff Calculation Engine (app.py)

```
def calculate_tangedco_domestic_bill(units):
if units <= 100:
    return 0
elif units <= 200:
    energy_charges = (units - 100) * 2.35
elif units <= 400:
    energy_charges = (100 * 2.35) + ((units - 200) * 4.70)
elif units <= 500:
    energy_charges = (100 * 2.35) + (200 * 4.70) + ((units - 400) * 6.30)
  else:
    energy_charges = (100 * 2.35) + (200 * 4.70) + (100 * 6.30) + ((units - 500) *
8.40)  electricity_duty = energy_charges * 0.05  fixed_surcharge = (units - 100) *
0.10
  return int(round(energy_charges + electricity_duty + fixed_surcharge))
```

Appendix B – Application Screenshots

The core operational user interface screens representing the end-to-end information discovery, telemetry simulation, hardware health diagnostics, and financial forecasting workflow of SolarSense are illustrated in Figure B.1 through Figure B.9 below.

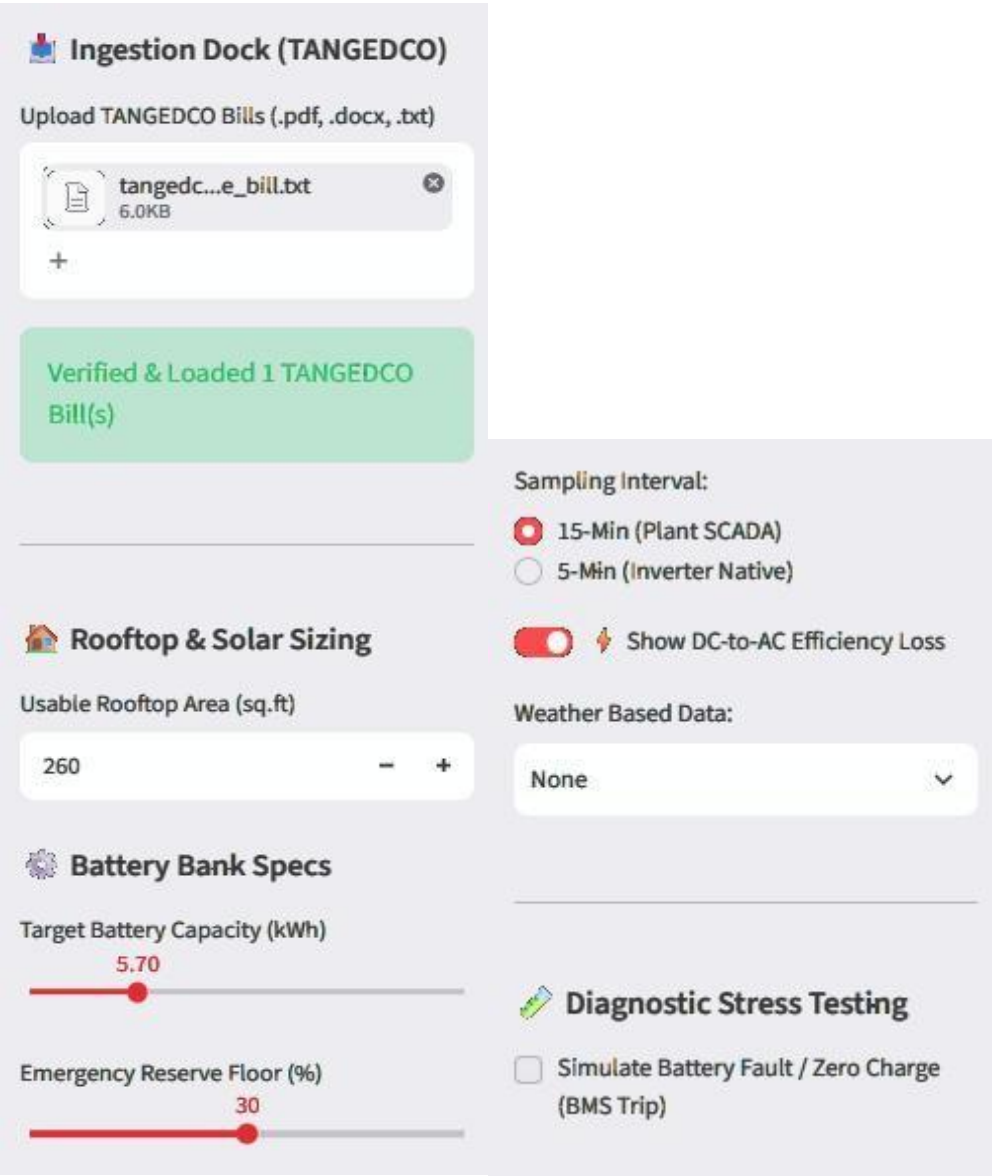


Figure B.1: Sidebar Ingestion Dock and Simulation Controls

Figure B.1 illustrates the SolarSense primary control sidebar. The **Ingestion Dock** validates and extracts data from uploaded electricity bills (PDF, DOCX, TXT). The **Configuration Panel** allows users to customize rooftop area, PV capacity, battery storage, and reserve cutoffs. It also provides controls to select SCADA intervals, load weather profiles, monitor inverter conversion losses, and inject BMS fault conditions for stress testing.



Figure B.2: Dual EB Predictor and Consumption Docket Ribbon

Figure B.2 displays the primary user dashboard upon ingesting an authentic TANGEDCO electricity bill. The Compact Docket Ribbon displays the consumer number, 60-day billing consumption (620 kWh), average daily demand (10.33 kWh/day), and dynamically derived rooftop array capacity (2.6 kW) and battery storage (5.7 kWh). The Dual EB predictor cards and pie gauges compare pre-solar baseline costs (₹3,195) with post-solar net payable charges (₹62), demonstrating a 98% bill reduction and active surplus export.

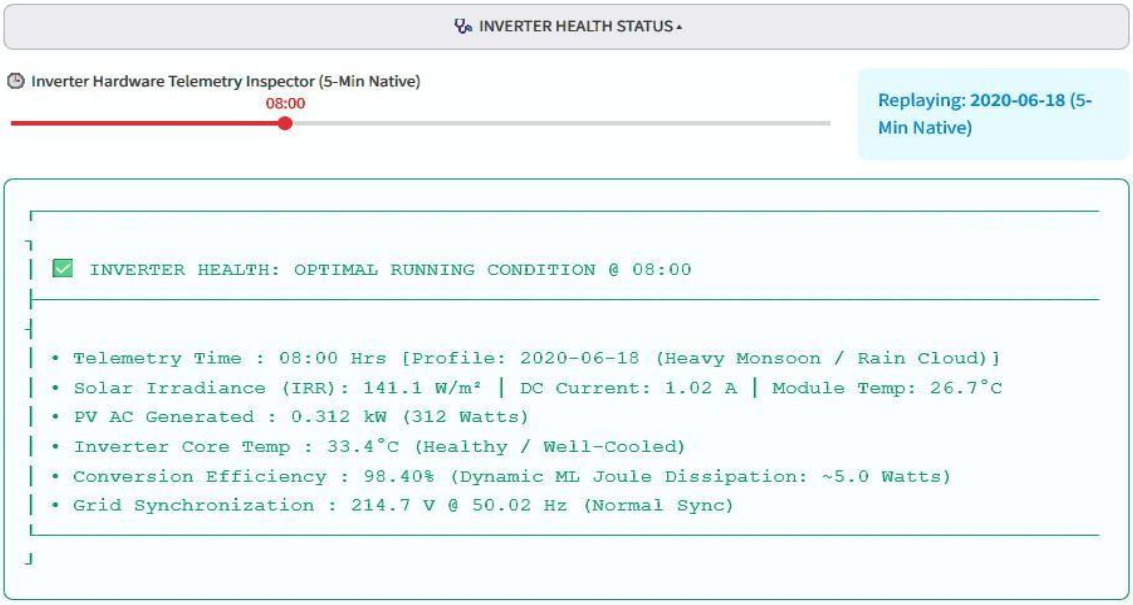


Figure B.3: Inverter Hardware Diagnostic Terminal and Health Telemetry

Figure B.3 illustrates the real-time inverter hardware diagnostic terminal operating on a 5minute sampling interval. It provides a simulated telemetry stream displaying live solar irradiance (W/m^2), direct current (A), module temperature (°C), AC power generated, dynamic conversion efficiency, and inverter core operating temperature to proactively detect thermal overload and islanded grid trips.



Figure B.4: Battery Storage Performance and Energy Flow Tracker

Figure B.4 demonstrates the 24-hour battery management and energy flow inspector. It visualizes instantaneous State of Charge (SoC %), net battery charge/discharge flow rates, and static transfer switch (STS) grid bypass statuses. The tracker verifies automatic power routing between self-consumption, battery reserve protection, and utility grid failover.

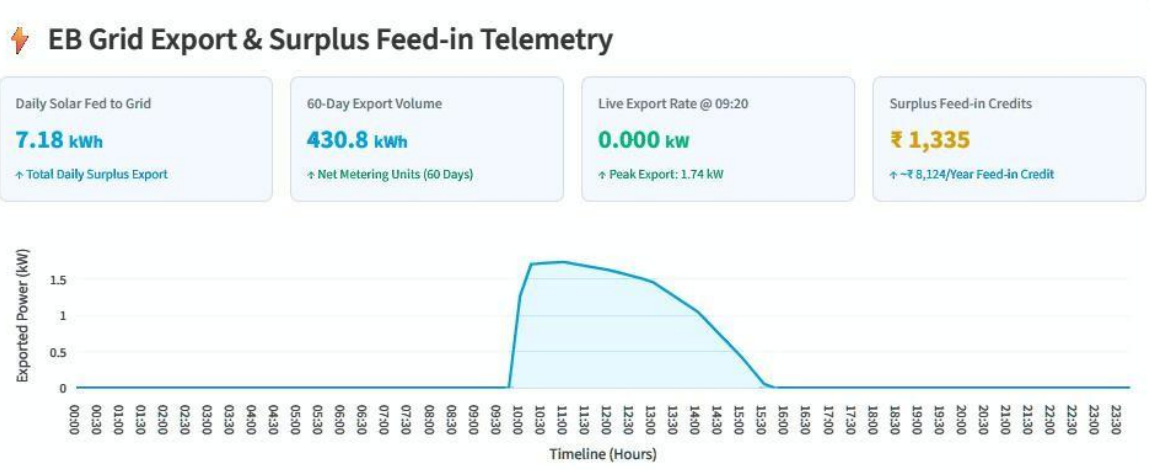


Figure B.5: EB Grid Export Telemetry and Surplus Feed-in Revenue Dashboard

Figure B.5 depicts the utility grid feed-in monitoring interface. It showcases daily net exported solar power, cumulative bimonthly net metering units, real-time export curves, and dynamic financial credits earned under official regulatory feed-in tariff rates.

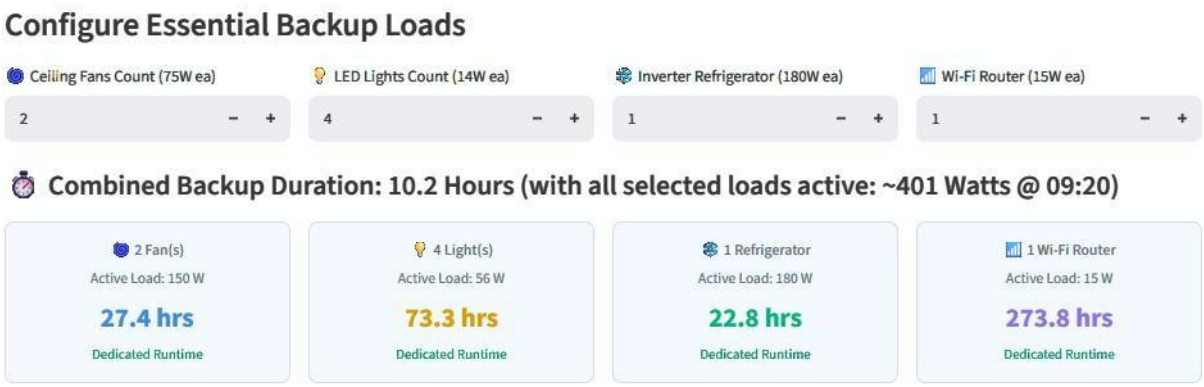


Figure B.6: Essential Backup Circuit Runtime Calculator

Figure B.6 shows the essential backup circuit configurator. Users can allocate reserve storage energy across critical domestic appliances—including ceiling fans, LED lighting, inverter refrigeration, and network routers—calculating dedicated and combined operational runtimes during simulated grid blackout events.

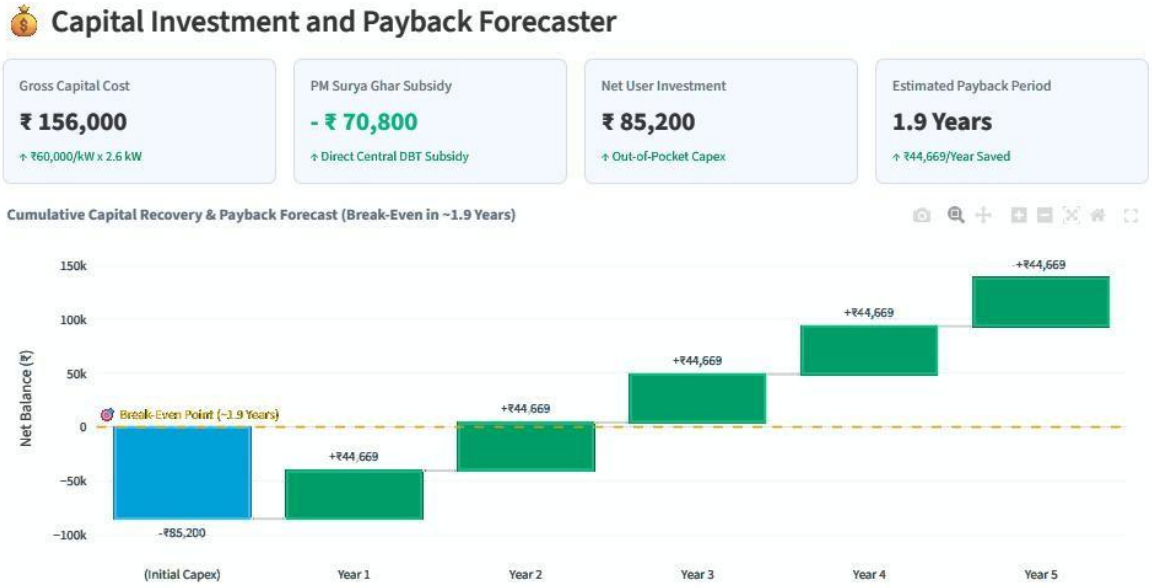


Figure B.7: Capital Investment Recovery and Payback Waterfall Forecaster

Figure B.7 presents the financial amortization and payback modeling dashboard. The waterfall chart maps initial gross capital outlay against central PM Surya Ghar direct benefit subsidies, illustrating annual electricity bill savings and projecting the exact break-even timeline for capital recovery.

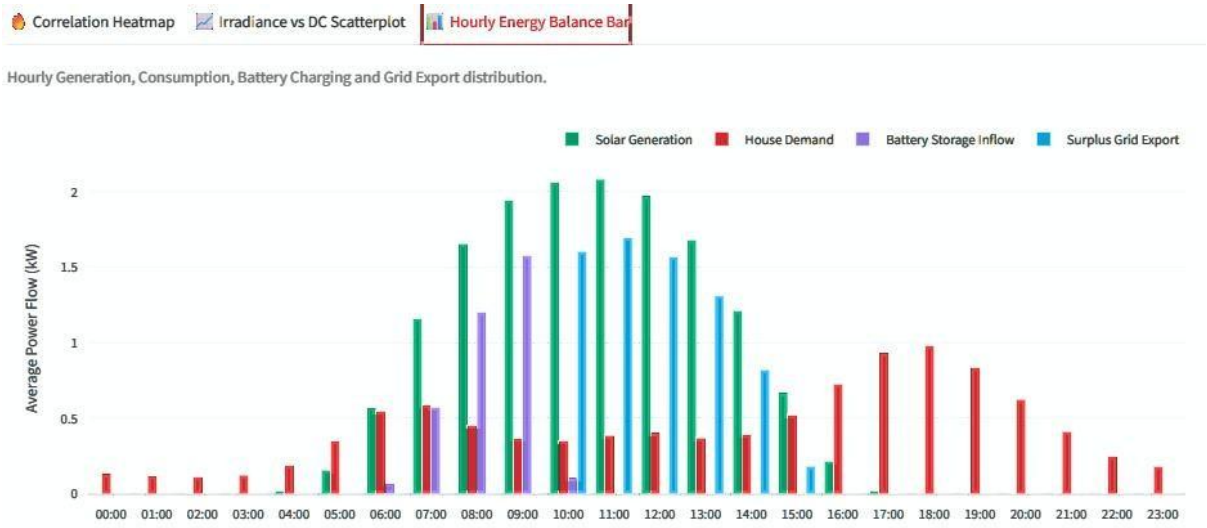


Figure B.8: 24-Hour Grouped Energy Balance Distribution

Figure B.8 illustrates the 24-hour grouped energy dispatch bar chart. The visualizer details the hour-by-hour operational distribution across solar generation, domestic household demand, battery charging absorption, and surplus utility grid exports.

Inspect Full 24-Hour Telemetry & Loss Ledger

Showing 96 records at 15-Min (Plant SCADA) native resolution.

	Timestamp	DC_Gen_kW	Inverter_Eff_%	AC_Dispatched_kW	Conversion_Wasted_W	House_Load_kW	Battery_Flow_kW	Battery_Stored_kWh	Bat
22	05:30	0.198	98.40%	0.193	3.1 W	0.375	0.000	1.52	
23	05:45	0.272	98.40%	0.265	4.2 W	0.431	0.000	1.52	
24	06:00	0.352	98.40%	0.344	5.5 W	0.485	0.000	1.52	
25	06:15	0.506	98.40%	0.494	7.9 W	0.525	0.000	1.52	
26	06:30	0.659	98.40%	0.643	10.3 W	0.566	0.078	1.54	
27	06:45	0.813	98.35%	0.793	13.1 W	0.606	0.187	1.57	
28	07:00	0.966	98.22%	0.942	16.8 W	0.639	0.303	1.64	
29	07:15	1.113	98.10%	1.086	20.6 W	0.604	0.483	1.74	
30	07:30	1.261	97.97%	1.230	25.0 W	0.568	0.662	1.89	
31	07:45	1.409	97.84%	1.374	29.7 W	0.533	0.841	2.09	
32	08:00	1.551	97.72%	1.513	34.5 W	0.498	1.015	2.33	
33	08:15	1.648	97.64%	1.607	37.9 W	0.468	1.139	2.60	

Figure B.9: High-Resolution SCADA Telemetry and System Loss Ledger

Figure B.9 exhibits the high-resolution tabular dispatch ledger. The data table logs complete time-series records of power generation, conversion dissipation losses, domestic consumption, battery state of charge, and net grid transactions at native plant SCADA intervals.

Appendix C – Sample Input and Output

A sample end-to-end interaction walkthrough is provided in Table C.1 below to demonstrate how SolarSense processes an uploaded TANGEDCO electricity bill, executes predictive machine learning inference, and generates actionable microgrid telemetry and financial results.

System Entity	Input Data	Processed Output
1. Bill Ingestion	Uploaded Document: `TANGEDCO_Bill_July2026.pdf`	Valid TANGEDCO LT-1A Domestic Bill Verified
2. Regex Parsing	Raw text stream of 60-day billing cycle	Consumer No: 03-024-004-1284 | Units: 720 kWh | Days: 60 | Baseline Bill: ₹3,840
3. System Sizing	Calculated Daily Demand: 12.00 kWh/day	Recommended PV Array: 3.0 kW | Storage: 6.6 kWh | Rooftop Area: 300 sq.ft
4. Weather Profile	Selected Date: `2020-05-15 (Peak Clear Summer)`	Peak Irradiance: 990 W/m² | Daily Macro Yield: 5.62 kWh/kW
5. Hardware Telemetry	12:45 Hrs Inverter Telemetry Point	Inverter Temp: 48.2°C (Healthy) | Efficiency: 98.15% | Grid Sync: 231.4V (Normal Sync)
6. 24h SCADA Dispatch	Diurnal Energy Flow Summation	Solar Dispatched: 16.80 kWh | Battery Losses: 0.182 kWh | Grid Export: 14.85 kWh
7. Financial Return	Gross Capex: ₹1,56,000 | PM Surya Ghar Subsidy: ₹70,800	Net Investment: ₹85,200| Annual Savings: ₹44,669 | Payback Period: 1.9 Years

Table C.1: Sample Ingested Bill and Telemetry Simulation Output Record